Gait Classification with Unsupervised k-means Clustering using Temporal and Spatial Characteristics of Plantar Pressure

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Abstract

This study investigated how effectively a pressure-sensor-embedded insole could differentiate between standard, antalgic, equinus, out-toed, and calcaneal gait patterns using temporal and spatial plantar-pressure characteristics with unsupervised *k*-means clustering. A custom insole containing six force-sensitive resistors was used to collect plantar-pressure data, from which five step-level characteristics were extracted and clustered. The resulting clusters achieved an Adjusted Rand Index (ARI) of 0.9080, correctly assigning 170 of 177 analyzed steps to their corresponding gait clusters. These results indicate that the extracted temporal and plantar-pressure characteristics contained sufficient information to distinguish the five recorded gait conditions under the tested experimental conditions.

Acknowledgments

I'd like to thank Professor Sarah Marzen for guiding me throughout this journey of writing my first research paper, Pioneer Academics for providing all of the resources and assistance I needed, and my parents for their unwavering support in anything I do.

Thank you all!

Contents

1 Introduction

Gait is the pattern of movement used during walking and can be affected by pain, injury, neurological conditions, and musculoskeletal conditions. Abnormal gait patterns can change how pressure is distributed across the foot and how long different parts of the foot remain in contact with the ground. Antalgic gait is commonly associated with pain and is characterized by a reduced stance time on the affected foot (Auerbach & Tadi, 2026).

Equinus gait is associated with excessive plantarflexion during walking (Robinson & Mielke, 2024). Out-toed gait involves an increased angle of the foot relative to the direction of travel (Khan et al., 2017). Calcaneal gait is characterized by excessive rearfoot loading and increased time spent on the rearfoot during the gait cycle (Sobel & Glockenberg, 1999). These differences in gait can potentially be identified by measuring the pressure applied to different regions of the foot during walking.

Gait can be analyzed using specialized systems that measure the movement and forces involved in walking. However, these systems can require specialized equipment and controlled environments, making them less practical for measurements outside of a laboratory. Plantar pressure measurements provide another method of analyzing gait by measuring how pressure is distributed across different regions of the foot during the gait cycle (Cavanagh et al., 1987). Pressure-sensing insoles allow these measurements to be collected while the subject is walking without requiring the same laboratory equipment. Previous research has demonstrated the use of sensors integrated into footwear for measuring plantar pressure during walking (Bamberg et al., 2008).

This study investigated whether a minimal pressure-sensing insole could differentiate between standard and simulated abnormal gait patterns using a small number of force-sensitive resistors (FSRs). The insole used six FSRs positioned under different regions of the foot and sampled pressure at approximately 24 Hz. The pressure data was processed to extract temporal and spatial characteristics from individual stance steps. These characteristics were then standardized and analyzed using unsupervised *k*-means clustering to determine whether the different recorded gait patterns formed distinct groups. The use of a small number of sensors and a relatively low sampling rate was intentional, allowing the system to be implemented using inexpensive and compact electronics rather than a specialized gait-analysis system.

Therefore, the research question investigated in this study was: How effectively can a pressure-sensor-embedded insole differentiate between standard, antalgic, equinus, out-toed, and calcaneal gaits by extracting spatial and temporal features and applying unsupervised *k*-means clustering

2 Background

2.1 Plantar Pressure and Gait

Plantar pressure is the pressure applied between the plantar surface of the foot and the ground during walking. The distribution of this pressure changes throughout the gait cycle as different regions of the foot contact and leave the ground. These changes can be used to characterize how loading is distributed across the foot during walking. Previous studies have measured plantar pressure across multiple regions of the foot to analyze differences in pressure distribution between gait conditions (Cavanagh et al., 1987). The foot can be divided into several regions for pressure analysis, including the rearfoot, midfoot, and forefoot, with these regions sometimes further divided into medial and lateral areas.

Hessert et al. (2005)used an insole containing 99 pressure sensors to measure plantar pressure across nine anatomical regions of the foot. The large number of sensors allowed pressure to be measured across a detailed representation of the plantar surface. However, a large sensor array also increases the amount of hardware and data required to collect and process the measurements. This provides a contrast to the six-sensor approach used in this study, where sensors were placed at selected regions of the foot rather than across the entire plantar surface.

2.2 Shoe-Integrated Pressure Sensing

Pressure sensors can also be integrated into footwear so that plantar pressure can be measured while a subject walks normally. Bamberg et al. developed a shoe-integrated wireless sensor system for gait analysis that used force-sensitive resistors (FSRs) positioned at several locations within the shoe (Bamberg et al., 2008). The system included sensors under the heel, first metatarsal head, and fifth metatarsal head, allowing pressure information from different regions of the foot to be collected during walking. The sensors were integrated into the shoe rather than requiring a stationary measurement platform, demonstrating that gait-related pressure data can be collected using sensors embedded within footwear.

The sensor placement used in this study follows a similar principle, with FSRs positioned under the heel, first metatarsal, fourth metatarsal, fifth metatarsal, medial arch, and lateral arch. Instead of attempting to measure pressure across the entire plantar surface, the sensors were positioned to provide measurements from the rearfoot, forefoot, and medial and lateral regions of the foot. This reduced the number of sensors required while still providing pressure measurements from multiple regions that could be used to calculate spatial characteristics.

2.3 Gait Characteristics and Classification

Plantar pressure measurements can contain both spatial and temporal information about gait. Spatial characteristics describe how pressure is distributed between different regions of the foot, while temporal characteristics describe when and for how long pressure is applied during the gait cycle. These characteristics can be used to identify differences between gait patterns because abnormal gait can alter both the distribution and timing of plantar pressure.

In this study, the pressure measurements were used to calculate the duration of the stance phase, the relative difference between heel and forefoot pressure, the relative difference between medial and lateral pressure, and the proportion of total pressure contributed by the heel and forefoot. These characteristics were selected to represent both the temporal duration of a step and the spatial distribution of pressure across different regions of the foot.

2.4 Clustering

Clustering is an unsupervised machine-learning method that groups observations based on similarities between their characteristics. Unlike supervised classification, clustering does not require the algorithm to be given the correct gait category for each observation during the clustering process. This makes clustering useful for determining whether different gait patterns naturally form separate groups based on their measured characteristics.

k-means clustering separates observations into a specified number of clusters by assigning each observation to the cluster with the nearest centroid and then updating the centroid based on the assigned observations. This process is repeated until the cluster assignments converge. Because the characteristics extracted from the insole have different units and ranges, the features were standardized before applying *k*-means so that characteristics with larger numerical values would not disproportionately affect the distance calculations.

Previous plantar-pressure studies have demonstrated that pressure measurements can be used to characterize gait, while shoe-integrated sensing has demonstrated that these measurements can be collected using sensors embedded within footwear. However, the use of a small number of pressure sensors to extract a limited set of spatial and temporal characteristics and then separate simulated gait patterns using unsupervised clustering provides a more constrained approach. This study investigates whether the information collected by a six-FSR insole is sufficient to distinguish between standard and simulated abnormal gait patterns using *k*-means clustering.

3 Methods

3.1 Insole Design

The pressure sensitive insole consists of an array of force sensitive resistors (FSRs), a custom 3D printed TPU 95A insole, and an Arduino Nano mounted on a custom soldered perfboard. The hardware revolves around three basic principles: a non-invasive design, easy integration and removal, and no cluttered wires or large processors. The sensor array consists of six FSRs, each located at a specific point of a men's size ten shoe insole. The more force is applied to the FSR, the less resistance is measured, so the FSR can give a relative estimate of the pressure being applied to it. Pressure is force per unit area, but because the force-sensitive area of each FSR is the same, changes in force applied to an FSR correspond proportionally to changes in pressure over that sensing area. Therefore, the FSR's output is treated as a relative measure of localized plantar pressure in this paper.

The 2 kg model of the DF9-16 was used as the FSR in the Insole, as shown in Figure 1. This model fits design principles one and two because of its rated thickness of less than 0.3 mm and has an added benefit of having a relatively low cost of 3 to 5 USD (Hanwei Electronics Group Corporation, 2025). Their pressure-sensitive diameter is about 7.8 mm, and each DF9-16 can read pressures from 20 g up to 2 kg. Although the human body weighs more than 2 kg, previous research shows that plantar load is not distributed uniformly across the foot. During barefoot standing, approximately 60% of the weight-bearing load was measured beneath the heel, 8% beneath the midfoot, and 28% beneath the forefoot, with minimal loading beneath the toes (Cavanagh et al., 1987).

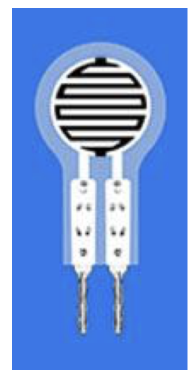

Figure 1: DF9-16 FSR

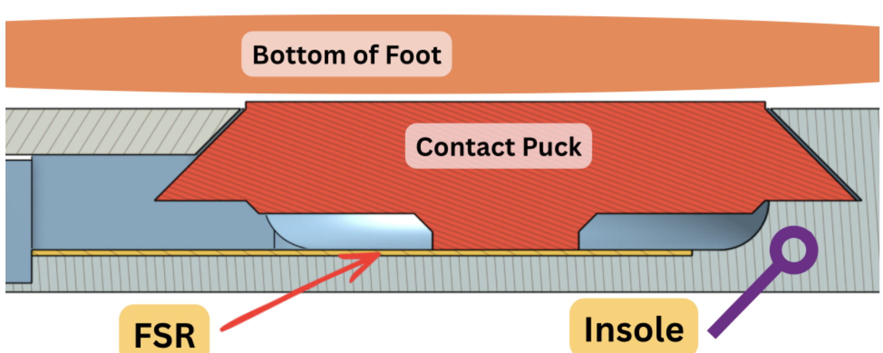

Figure 2: Contact Puck Cross-Section

If the wearer weighs 50 kg, approximately 30 kg of body weight would be supported by the anatomical heel region under the standing conditions measured by Cavanagh et al. (1987). However, this does not mean the heel FSR has to measure 30 kg. The Insole's design has a contact puck between the FSR and the sole, as shown in Figure 2. The contact puck acts as a buffer to protect the FSR from debris and damage while also transferring the sole's load. With a diameter of 14.2 mm, it only allows the sensor to measure a small area of the heel. Furthermore, a voltage divider was used to reduce the saturation of the FSRs so they could sense more force. The specifics will be documented in Electronics, but in the recordings, the maximum sensor ADC reading was 212 out of 1023, corresponding to 20.7% of the microcontroller's full range. This corresponds to an FSR resistance of about 3.8kΩ, which is well within the FSR's measurable range and far from the voltage-divider's saturation point, so no saturation was observed in the FSR's readings.

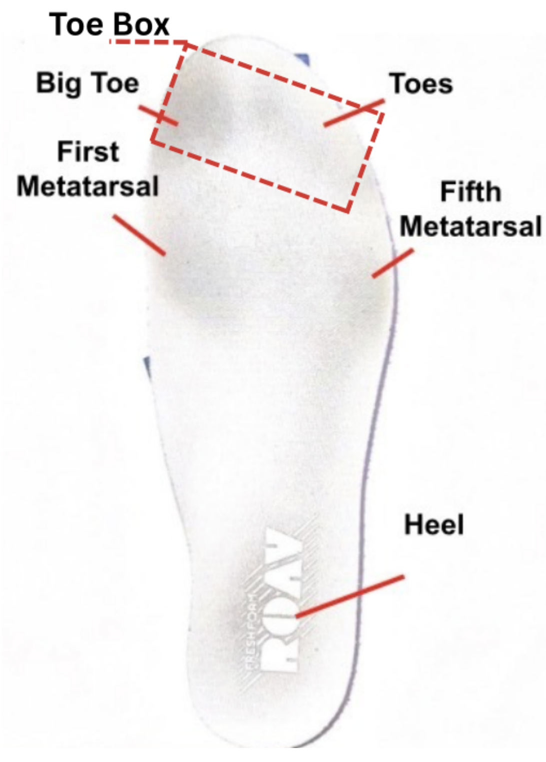

Figure 3: Foam Insole

Figure 4: Insole V1 Prototypes
Left – FSR Location Draft
Right – Insole V1 (rigid)

The 3D printed Insole's shape is based on the stock men's 10 New Balance Fresh Foam Roav. In Figure 3, black marks can be seen near the toe box, first metatarsal, fifth metatarsal, and heel. The marks were created by months of pressure and friction from the foot against the foam surface, and they highlight where the foot most often contacts the ground. The Insole's original design had three FSRs on the heel, first metatarsal, and fifth metatarsal, as shown on the left of Figure 4. These locations were chosen because they had the most wear on the foam insole; the toe box was excluded because the toes could move independently and give unreliable readings. In a previous study, Bamberg et al. (2008) incorporated force-sensitive resistors beneath the heel, first metatarsal head, and fifth metatarsal head in their shoe-integrated gait-analysis system (Bamberg et al., 2008).

As testing started for Insole V1, the data collected during recordings were graphed in Google Sheets with time as the x-axis and the plantar pressure data as the y-axis. Each step was easily identifiable due to the swing phase of a standard walking gait causing zeros to appear in the pressure data. However once different gaits started to be tested, such as out-toed (duck-walk), there weren't any easily identifiable characteristics between them. In Figure 5, three different gaits were recorded and graphed: walking forwards, walking backwards, and out-toed/duckwalk forwards. Even though each gait looks different from the outside, with a purely data perspective, each gait is the same.

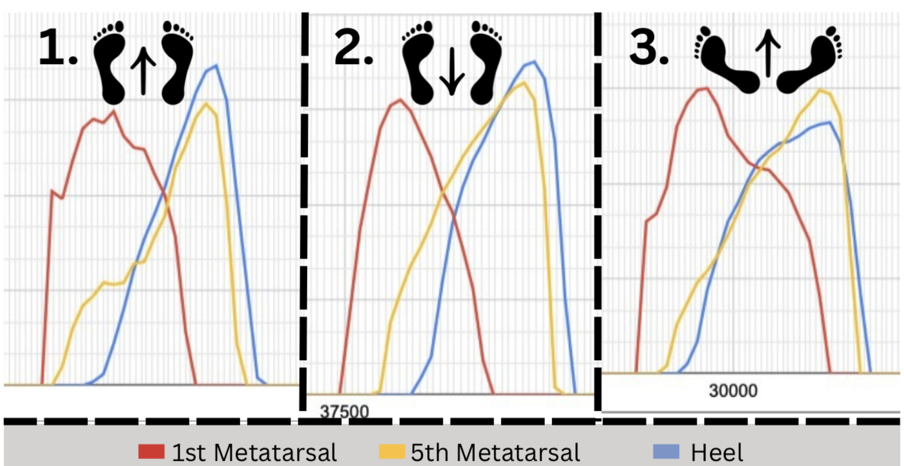

Figure 5: Data from Insole V1
Left – Walking Forwards
Middle – Walking Backwards
Right – Walking Out-toed/Duck-walk

To differentiate between multiple gaits, more sensors were needed in the Insole; specifically ones along the medial and lateral arches of the foot. The arches are one of the most important parts of the foot to monitor, because when rolling an ankle, more force shifts to the lateral longitudinal arch and the ankle's tendons are strained. (CITE) Additionally, when balancing on one foot, the body often corrects balance by shifting weight from medial to lateral arch—the foot's roll axis. (CITE) At first, seven sensors were considered for the new design, adding four FSRs—two for the forefoot on metatarsals 2 and 4, and one along each arch, highlighted with green in Figure 6. However, the wiring plan quickly became too complex and too messy, violating principle three. For seven FSRs, fourteen wires must be used—two for each sensor, and each must be routed carefully to avoid overlaps and assembly issues. Multiple wire channel designs were considered, but fourteen wires were too much.

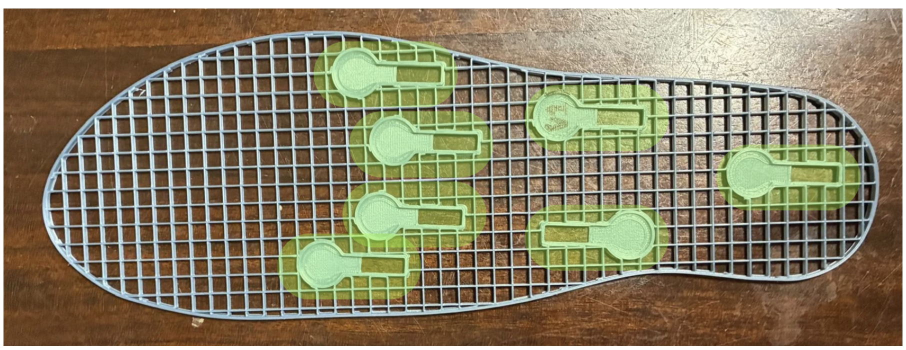

Figure 6: Seven Sensor Design

Reducing the number to six FSRs and modifying the sensor orientation helped a little, but the biggest difference was how the wires were arranged. A FSR has two non-polarized terminals. One of them must be connected to a voltage source, such as five volts, and the other is connected to a microcontroller's analog input using a voltage divider. The analog inputs must be separated, but the five volt rail can be linked in parallel. Using this logic, the twelve FSR wires for six sensors can be reduced to seven wires coming out of the Insole. As shown in Figure 7, two main channels of wires diverge into separate branches, similar to how a river divides into multiple tributaries. The five volt wires are spliced into a Y shape to distribute the voltage in parallel across the interconnected FSR arrays.

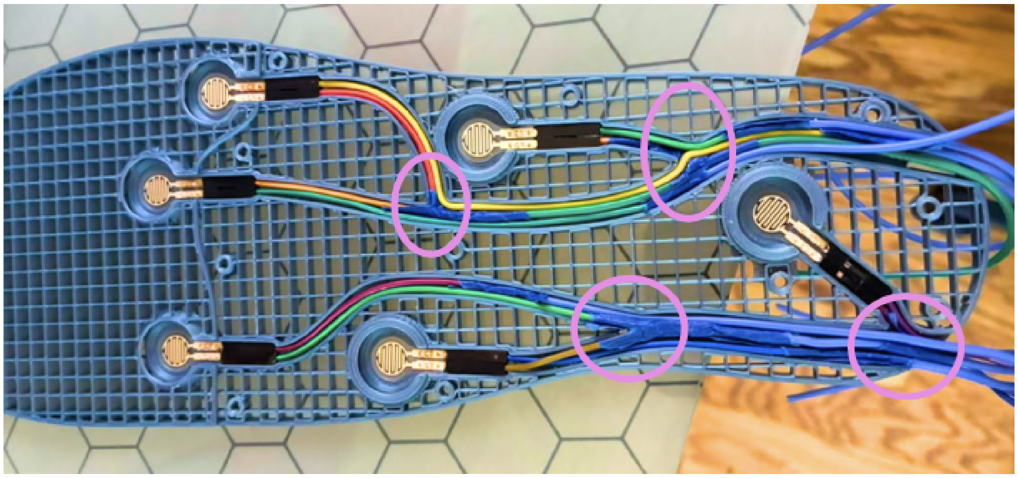

Figure 7: Insole V3 Wire Channels

Each FSR unit is nestled into a tight-tolerance cavity in the Insole. Since the sensor is so thin, measured at 0.15 mm, there is no way to screw it in, and glue or tape can make a bump underneath the FSR, causing inaccurate and imprecise measurements. As shown on the left in Figure 8, the contact puck on top of the sensor is locked in place with a 45° chamfer, ensuring it won't come loose. With an FSR underneath it, it effectively prevents the FSR from bending, and the cavity helps secure the FSR from shifting. To connect the FSR to the wires, a different approach was needed. Typically, hot solder would be applied to the exposed metal on two wires, thereby permanently fixing them together. The downside to this approach is that the FSRs are heat-sensitive and the high temperature of the soldering iron risks damaging the sensor (JIMBLOM & BBOYHO, 2016). According to Sparkfun Electronics, the Amphenol FCI Clincher Connector is suggested to securely connect wires to an FSR without soldering (JIMBLOM & BBOYHO, 2016). Even so, standard Dupont connectors (Right of Figure 8) were chosen because of their narrow profile, availability, and simple shape for integration into the 3D printed Insole, and because the FCI Clincher Connectors had a wide shape that would take up valuable space inside the Insole and hinder its ability to bend well.

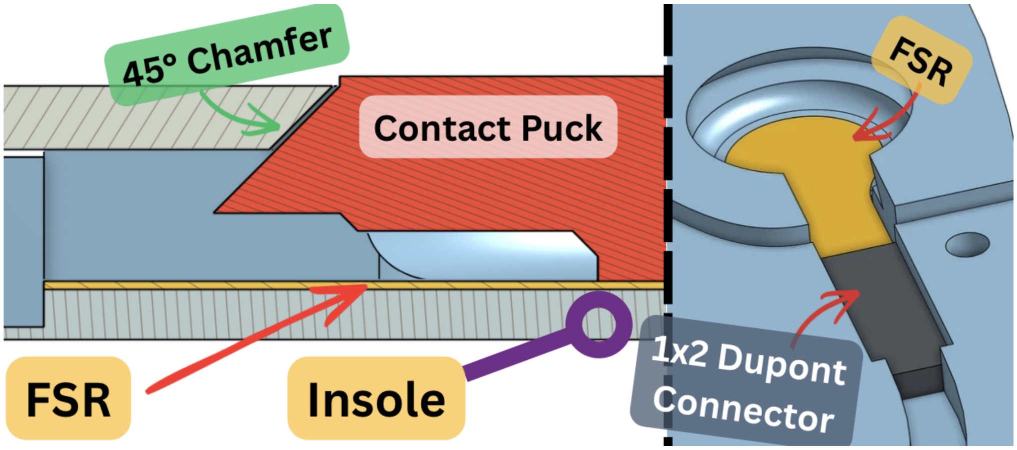

Figure 8: Insole V3 FSR Cavity
Left – Contact Puck Chamfer
Right – Dupont Cavity

The final Insole design uses six FSRs: three in the forefoot, two in the arches, and one on the heel. As shown in Figure 9 anatomical locations are: first, fourth, and fifth metatarsals, the heel, and along the medial and lateral longitudinal arches.

Figure 9: Final Insole Design

TPU95A is a flexible material that can be 3D printed, and it was used to print the Insole. Since the Insole is large and flat, printing the smooth top and bottom layers can take up to three times as long. For prototyping's sake, the top and bottom layers were removed, making the structural pattern - infill - visible. The type of infill is very important in the Insole's design, as it dictates its direction and amount of flexibility and compliance (MonroeScarlett, 2026).

Figure 10 shows gyroid infill, a three-dimensional, repeating lattice structure used inside a 3D-printed part to reduce material usage and weight while maintaining structural strength. Its continuous, interconnected geometry distributes applied forces throughout the print (Delfosse, 2025). Twenty-five percent gyroid infill was used in the final Insole because it offers equal flexibility in all three axes and reduced both print time and used material.

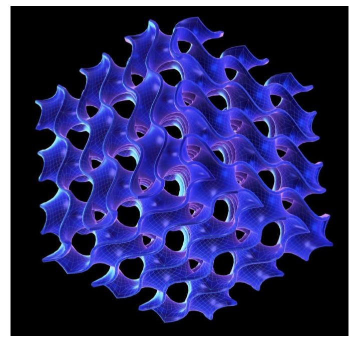

Figure 10: Digital Model of a Gyroid Structure.
Note. Adapted from (Delfosse, 2025).

The Insole has a cover piece which is screwed into the main Insole, and it's used to reduce tangling by securing the wires in their own respective channels, shown in Figure 11. The cover also functions as an adjustable inhibitor to control the amount of flexibility in the wire channels and midfoot to ensure the Insole isn't too soft, because according to Lee (2025), that can cause overpronation, a condition of excessive inward rolling of the feet, making the recorded data unreliable.

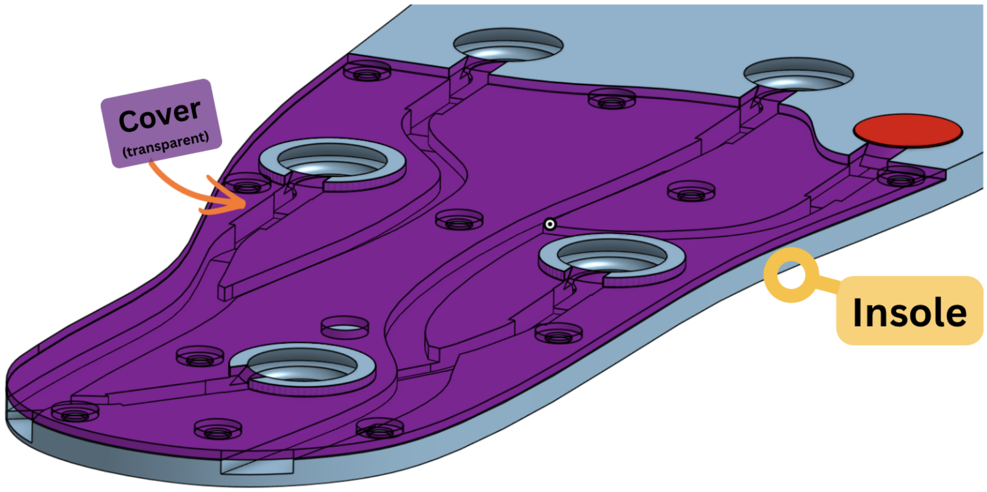

Figure 11: Insole Cover

To log all of the data, the electronics must be mounted somewhere close to the Insole. A model from the 3D printing model database and hub Printables inspired the electronics mount. The Shoe Horn by @fifindr is "A compact shoe horn designed to help slide your foot into a shoe more easily" (fifindr, 2026). The electronics mount, shown in Figure 12 uses a shoe horn-shaped clip to be press-fit onto almost any shoe. The microcontroller, micro SD card module, and battery are all mounted onto it using screws, and the eight wires are nestled behind the curved plate between the wearer's heel and the back of the shoe to ensure no wires are put under tension.

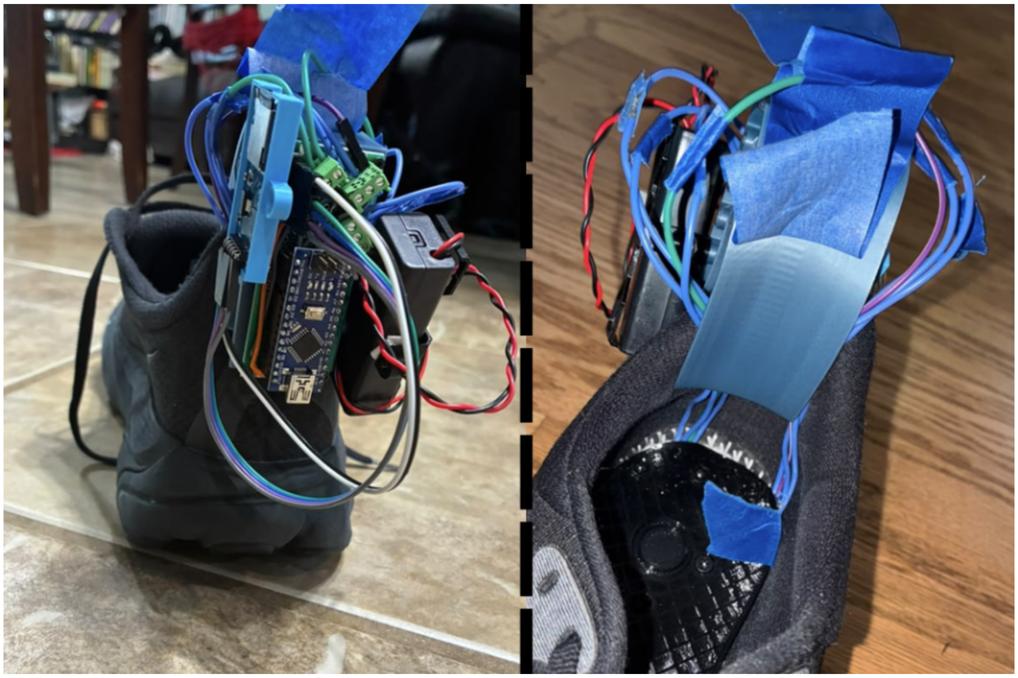

Figure 12: Electronics Mount

3.2 Electronics

As mentioned in Insole Design, a voltage divider was needed to convert the FSR's changing resistance into a measurable voltage that could be read by the Arduino Nano's analog pins. The `analogRead()` function converts the input voltage into a corresponding 10-bit ADC value. (Singh, 2024). For the Nano's default five volt reference, the ADC produces values from 0 to 1023 as the input voltage increases from zero volts to approximately five volts. According to JIMBLOM (n.d.), "A voltage divider is a simple circuit which turns a large voltage into a smaller one". It has two resistors connected in series, with the output voltage at their intersection determined by their relative resistances. Using Equation 1, the output voltage can be easily calculated.

$$V_{\text{out}} = V_{\text{in}} \left(\frac{R_2}{R_1 + R_2} \right) \tag{1}$$

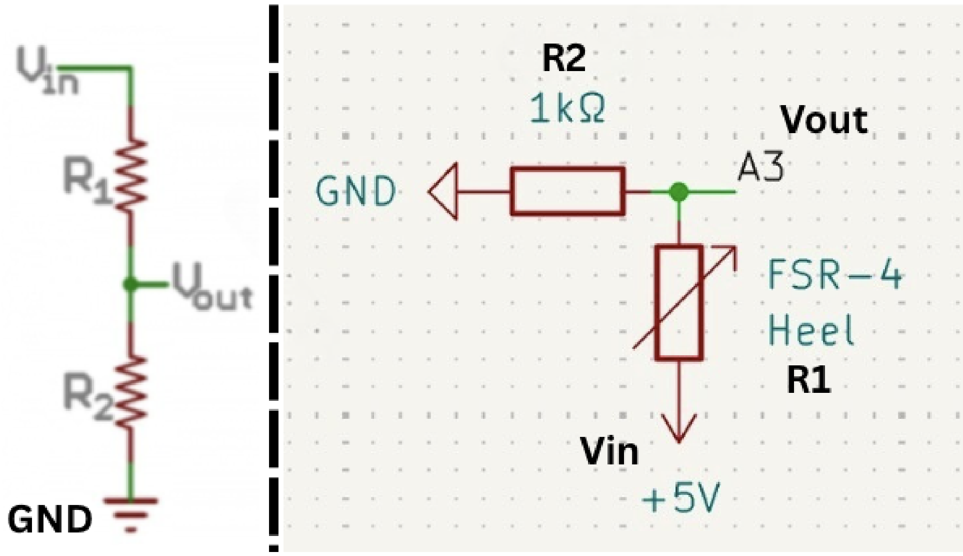

Figure 13: FSR Voltage Divider
Left – Examples of Voltage Divider Schematics
Note. From *Voltage Dividers*, by JIMBLOM (n.d.).
Right – Single Insole FSR Circuit Schematic
Made in KiCAD

Another factor mentioned in Insole Design, the force on an FSR is inversely proportional to the resulting resistance. With this principle in mind, a voltage divider can be constructed with the FSR as R_1 and a fixed resistor as R_2, shown as a circuit diagram in Figure 13. The FSR acts as a variable resistor, so when its resistance decreases due to an increase in applied force, the output voltage, V_{out}, will increase, causing the value of `analogRead()` to increase as well. For the fixed resistor, the value of 1kΩ was chosen during initial testing because it produced less saturation at higher applied forces than the 10kΩ resistor initially considered.

3.3 Data Transfer

	A	B	C	D	E	F	G
1	Time_ms	5th_Metatarsal	4th_Metatarsal	1st_Metatarsal	Heel	Medial_Arch	Lateral_Arch
2	1745	0	0	0	0	0	0
3	1786	0	0	0	0	0	0
4	1827	0	0	0	0	0	0
5	1869	0	0	0	0	0	0
6	1910	0	0	0	0	0	0
7	1951	0	0	0	0	0	0
8	1993	0	0	0	0	0	0
9	2034	0	0	0	0	0	0
10	2075	0	0	0	0	0	0
11	2117	0	0	0	0	0	0
12	2158	0	0	0	0	0	0

Figure 14: Sample of .CSV data from LOG7 in Google Sheets

Each LOG.CSV is filled with plantar pressure data from the Insole. As shown in Figure 14, the six FSR values are separated into six columns, with time as a seventh variable. Everything is saved into the micro SD card as separate LOG.CSV files, where each time the Arduino Nano turns on, it scans the files in the SD card and identifies the next available LOG number. For example, if LOG1, 2, 3, and 4.CSVs exist already, the program will automatically see that, make a new file called LOG5.CSV, and start logging pressure data into it. After all of the necessary LOGs and gaits are collected, the data has to be transferred from the micro SD card to the Python programs for analysis. As mentioned in Electronics, a bluetooth module was considered as an add-on so the data could be immediately transferred from the Arduino Nano to a phone or laptop. However, the Nano only has 2 KB of RAM (Arduino, 2026), and adding a bluetooth module would introduce additional memory and processing requirements. There is a version of the Nano that has bluetooth integrated into the board, but adding wireless transmission would introduce additional data-transfer requirements that were unnecessary for the intended sampling rate. As shown in Figure 15, the standard HC-05 Bluetooth module is nearly the same size as the Arduino Nano, but lacks mounting holes, making it difficult to integrate into the electronics mount without increasing its size. Although using the micro SD card required manually transferring the data, it doesn't introduce additional requirements for its function, and even though the SD read/write module is much bigger than the HC-05, it has mounting holes which allow the creation of a mount that can be screwed into the electronics mount.

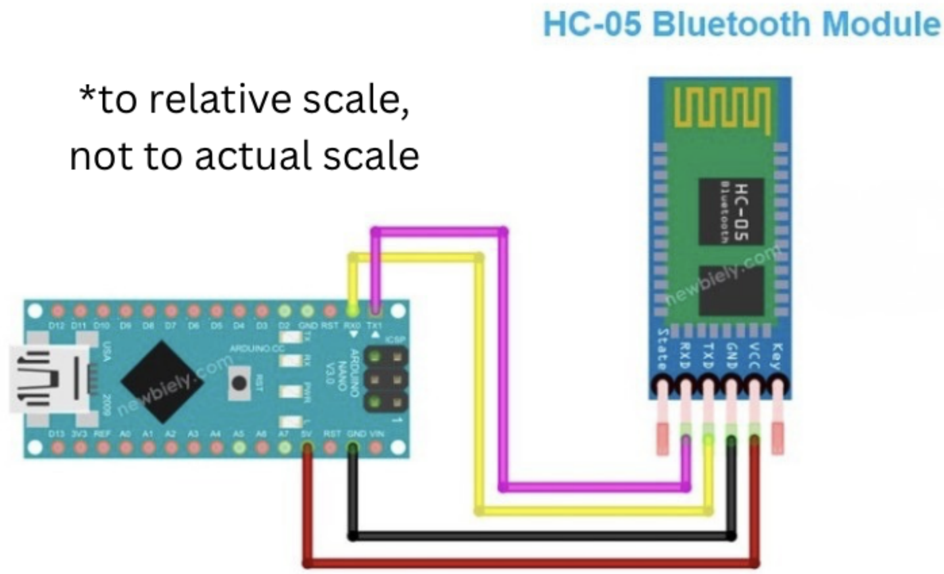

Figure 15: HC-05 Bluetooth module and Arduino Nano
Note. From *Arduino Nano - Bluetooth*, by Newbiely (2026).

3.4 Data Collection

Typically to collect foot pressure data for five gaits, four of which can cause injury, five subjects with each gait condition and similar body characteristics would normally be used. However, finding the five subjects would require looking at hundreds, if not thousands, of candidates. Then, an informed consent form would have to be signed and approved by an Institutional Review Board or Independent Ethics Committee (Gupta, 2013). Without any access to that, I decided to be the only subject from which data would be collected, and the five gait conditions would have to be simulated to the best of my ability. As the subject, a 183 centimeter, 56.83 kilogram Chinese male with a 101.6 centimeter hip height, I have to try and keep the same constants in each data collection walk. The men's 10.5 Brooks Hyperion 2 and the Insole were used to collect foot pressure data for each of the five gaits. Each .CSV file of data contains a LOG-specific gait, so cleaning and filtering the foot pressures are easier. The interval at which data is recorded is always 24.10 hertz (every 41.5 ms). As another constant, the data for each LOG was collected on the same surface hardness, and the walking surface and cadence was maintained throughout the recording of a LOG to get the most consistent step data.

3.4.1 Standard Gait

Standard Gait is the gait the subject walks everyday. It won't be called a normal gait because that's relative to the person who is walking. There are too many factors, such as leg length, weight, fitness, foot size, and many more, that can impact and alter the person's everyday walk. Collected as LOG7.CSV, it contains seventeen minutes and thirty-one seconds of walking using a gait without issues to collect as much data as possible, and it contains 863 raw steps (850 after filtering). As shown in Figure 16, the route was tracked using an Apple Watch SE (1st Generation, GPS + Cellular) and went in a loop around the subject's neighborhood. Even though the route was recorded for thirty-one minutes with a distance of 1.88 kilometers, LOG7 only collected seventeen minutes and thirty-one seconds of data over 1.22 kilometers at an average pace of fifteen minutes and twenty-four seconds per kilometer. This was due to heel whip in the subject's walking gait, which caused the microcontroller unit to be impacted and unplugged the SD card module, shown in Figure 17. According to Souza et al. (2015), a heel whip is defined as the medial or lateral rotation of the foot in the transverse plane during the initial swing phase.

Figure 16: LOG7 Route

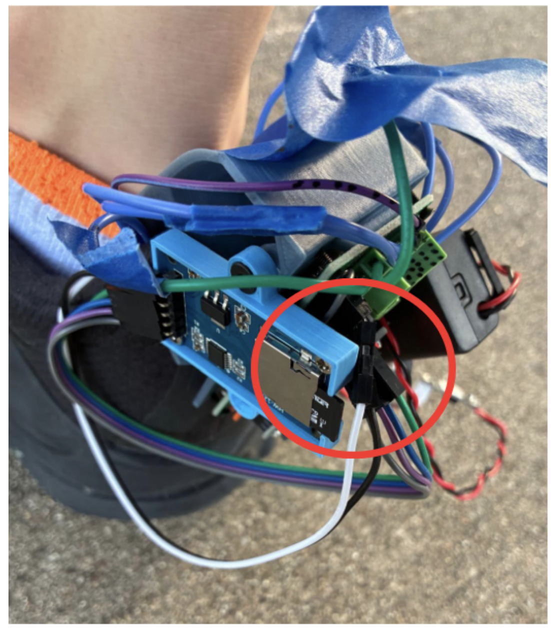

Figure 17: SD Card Impact

3.4.2 Limping Gait

Antalgic Gait, or a Limping Gait, is an abnormal walking pattern caused by pain that results in a limp, with the stance phase shortened relative to the swing phase (Auerbach & Tadi, 2026). It was collected as LOG11.CSV and contains 40 raw steps (38 after filtering) of a simulated antalgic gait where the subject, wearing the Insole on their right foot, shortened their stance on the ground and made their right foot spend less time on the ground.

3.4.3 Tiptoeing Gait

Equinus Gait, or a Tiptoe Gait, is an abnormal walking pattern characterized by plantar flexion of the ankle during both the stance and swing phases of gait (Robinson & Mielke, 2024). It was collected as LOG14.CSV and contains 38 raw steps (37 after filtering) of a simulated equinus gait where the subject maintained a flexed ankle whilst walking.

3.4.4 Duck-walking Gait

Out-toed Gait, or a Duck-footed Gait, is a walking pattern in which the foot is externally rotated relative to the direction of walking, resulting in an increased foot progression angle (FPA) (Khan et al., 2017). It was collected as LOG12.CSV and contains 27 raw steps (25 after filtering) of a simulated out-toed gait where the subject rotated both of their feet so it made an almost 160° angle whilst walking.

3.4.5 Heel-walking Gait

Heel-walking and calcaneal gait are different, because calcaneal gait has several other factors that are difficult to simulate. The heel-walking condition was simulated by maintaining increased rearfoot loading throughout the gait cycle. This condition resembles the excessive rearfoot loading characteristic of calcaneal gait, which is described as a shuffling, apropulsive gait with excessive time spent on the rearfoot during the gait cycle (Sobel & Glockenberg, 1999). This gait was collected as LOG13.CSV and contains 39 raw steps (37 after filtering) of the simulated calcaneal gait.

3.5 Data Filtration

After the data is transferred to a computer, the data has to be processed before being clustered by *k*-means. Each LOG contains plantar pressure data that has to be separated into individual steps before the data can be processed. The isolation algorithm identified the stance and swing phases of each recording by classifying a row as part of the swing phase when all six FSR readings were zero, indicating that there was no pressure being applied to the Insole. Conversely, a row was classified as part of the stance phase when at least one FSR recorded pressure. Each row in one category, stance or swing, was appended into an array consisting of other consecutive rows in the same category. The variable `Time_ms` would have kept on increasing throughout the data, so after each step is isolated, the time it starts is subtracted from each row's `Time_ms` column, ensuring each step starts counting time from zero. The step array was then numbered and put into either the `stance_step_dict` or the `swing_step_dict` dictionary.

The resulting `stance_step_dict` contained several unusually long stance detections that did not represent normal walking steps. For example, one stance segment in LOG7 contained 451 rows, corresponding to approximately 18.7 seconds of continuous pressure. A step this long is inconsistent with an individual walking step and was likely caused by the subject remaining stationary while the foot continued to apply pressure to the Insole. These outliers could skew the *k*-means algorithm and have to be removed.

A two-pass standard-deviation-based filtering method was used to remove abnormal stance detections. In the first pass, the number of rows in each stance step was calculated, and an upper threshold was established using the initial mean plus 0.2 standard deviations. Stance steps exceeding this threshold were excluded from the walking dataset. The mean and standard deviation were then recalculated using the remaining steps to establish a baseline for normal walking. In the second pass, steps falling more than 2.9 standard deviations below this baseline were removed as potential short-step glitches or noise. The remaining stance steps were kept for further processing.

The 0.2 standard-deviation multiplier was selected based on the observed separation between normal walking steps and unusually long stance detections. The resulting threshold consistently separated the long detections from the primary group of walking steps. To determine whether the selected multiplier substantially affected the resulting dataset, different upper thresholds were tested. For LOG11 through LOG14, thresholds spanning approximately 20 to 60 rows produced the same retained steps, showing that the separation between normal walking steps and the long detections was large enough that the exact threshold had little effect on the resulting dataset. LOG7 had the smallest separation, with the longest retained walking steps reaching approximately 18 rows and the nearest removed step occurring at approximately 26 rows. Therefore, LOG7 represented the most restrictive case for the selected threshold. Figure 18 shows the upper thresholds and unusually long steps for each gait type as red dashed lines and points, respectively. The lower thresholds and unusually short steps are shown as purple dashed lines and points, respectively, while the remaining filtered steps are shown as blue points.

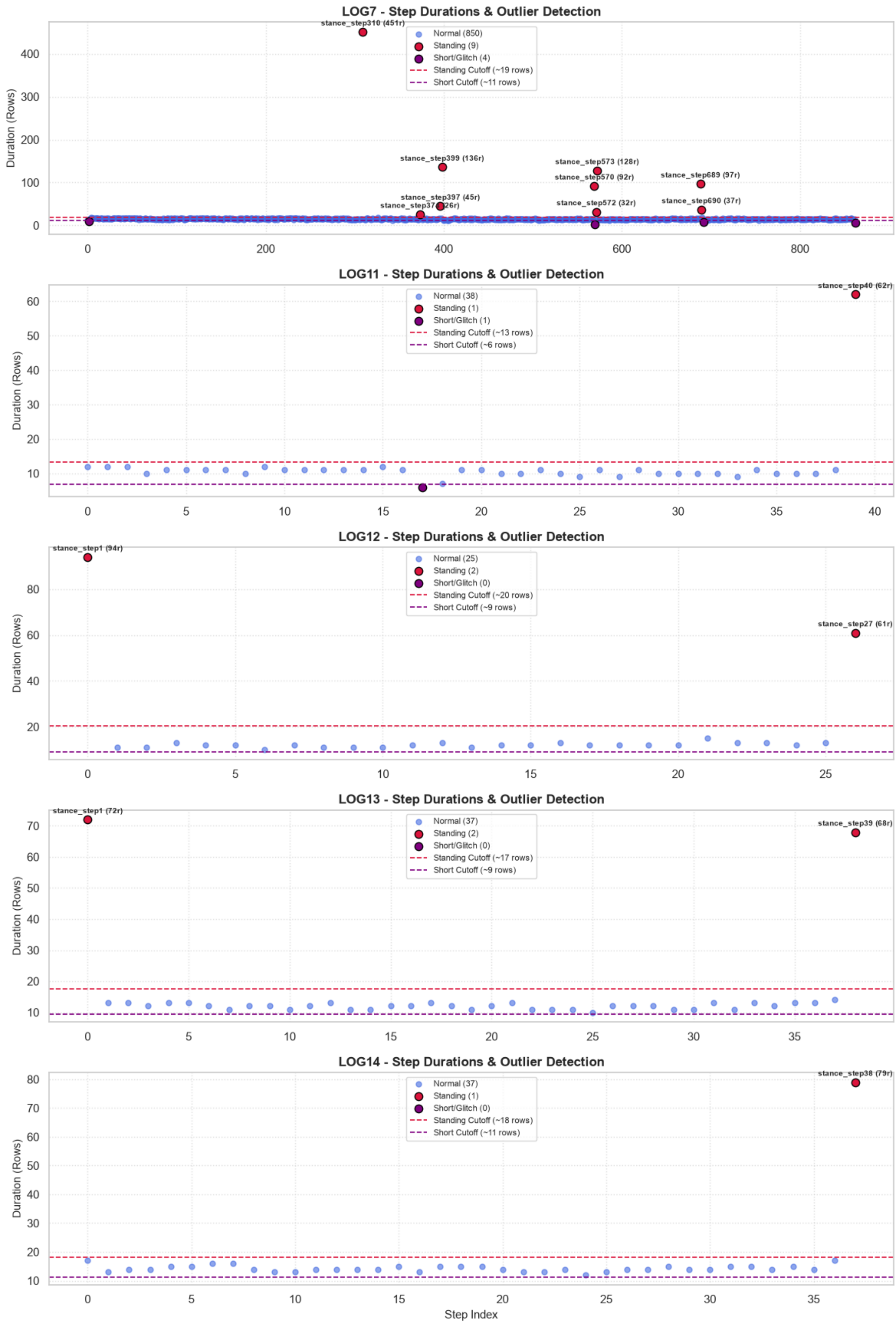

Figure 18: Duration vs Step Index Scatterplot of the 5 Gait Types

The filtering procedure was conservative, retaining 987 of the 1,007 raw stance steps, or approximately 98%, for analysis. The second filtering pass removed only 5 of the 992 steps remaining after the first pass, or approximately 0.5%. This shows that the lower threshold removed only a small number of potential short-step glitches while leaving the majority of the detected walking steps unchanged.

3.6 Data Processing

After filtering the five LOG.CSVs to remove outliers and short, glitched steps, each had their own filtered step count. LOG7 was the longest recording session out of all five, so it had a filtered step count of 850. LOG11, 12, 13, and 14 were shorter sessions, and so they had 38, 25, 37, and 37 steps, respectively. Because LOG7 contained a lot more steps than the other gait recordings (850 versus 25 to 38 steps), forty steps were randomly subsampled before clustering. This prevented the Standard Gait recording from disproportionately influencing the *k*-means model's result due to its much larger number of observations. From each LOG, five step characteristics were extracted from each step: duration, heel-forefoot index, medial-lateral index, heel percent of total pressure, and forefoot percent of total pressure.

Duration is a temporal characteristic that measures how long the foot remains in the stance phase. It's calculated using Equation 2, where t_{end} is the time in milliseconds when the step ends and t_{start} is the time in milliseconds when the step starts.

$$D = t_{\text{end}} - t_{\text{start}} \tag{2}$$

Both the heel-forefoot and medial-lateral indexes are on a scale of -1 to 1. The heel-forefoot index is calculated using Equation 3. Values closer to 1 indicate greater relative pressure on the heel, while values closer to -1 indicate greater relative pressure on the forefoot. A value near zero indicates that the heel and forefoot had relatively similar pressure. The medial-lateral index is calculated using Equation 4 and follows the same principle, with values closer to 1 indicating greater relative pressure on the medial side of the foot and values closer to -1 indicating greater relative pressure on the lateral side. A value near zero indicates relatively similar pressure between the medial and lateral regions.

$$I_{\text{HF}} = \frac{P_{\text{heel}} - P_{\text{forefoot}}}{P_{\text{heel}} + P_{\text{forefoot}}} \tag{3}$$

$$I_{\text{ML}} = \frac{P_{\text{medial}} - P_{\text{lateral}}}{P_{\text{medial}} + P_{\text{lateral}}} \tag{4}$$

where P_{medial} represents the summed pressure recorded by the first metatarsal and medial arch sensors, and P_{lateral} represents the summed pressure recorded by the fifth metatarsal and lateral arch sensors. P_{heel} represents the summed pressure recorded by the heel sensor, while P_{forefoot} represents the summed pressure recorded by the first, fourth, and fifth metatarsal sensors. The pressure values are summed across all samples within the stance step.

The heel percent and forefoot percent represent the proportion of the total recorded pressure contributed by each region. The heel percent is calculated using Equation 5, which divides the summed pressure from the heel sensor by the total pressure recorded by all six FSRs. Similarly, the forefoot percent is calculated using Equation 6, which divides the summed pressure from the first, fourth, and fifth metatarsal sensors by the total pressure recorded by all six FSRs. Unlike the two indexes, these values range from zero to one and describe the relative contribution of each region to the total recorded pressure.

$$P_{\text{heel\%}} = \frac{P_{\text{heel}}}{P_{\text{total}}} \tag{5}$$

$$P_{\text{forefoot\%}} = \frac{P_{\text{forefoot}}}{P_{\text{total}}} \tag{6}$$

where P_{total} represents the total pressure recorded by all six FSR sensors across the stance step.

These five characteristics were combined into a five-dimensional feature vector for each stance step. The resulting feature matrix therefore contained one row for each step and five columns corresponding to duration, heel-forefoot index, medial-lateral index, heel percent, and forefoot percent. Before applying k-means, the five characteristics were standardized using `StandardScaler` so that characteristics with larger numerical ranges, such as duration, would not disproportionately affect the clustering. The appropriate number of clusters was then determined using elbow and silhouette analyses.

To determine the appropriate number of clusters, an elbow analysis and silhouette analysis were performed for values of k from 2 to 10. The elbow method evaluates the inertia, or within-cluster sum of squared distances, for each value of k. As shown in Figure 19, the largest decreases in inertia occurred between $k = 2$ and $k = 4$, after which the decrease became smaller. This suggested that increasing the number of clusters beyond approximately four provided diminishing reductions in within-cluster variation. However, the elbow method does not provide a single objective selection of k, so silhouette analysis was also used.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{7}$$

The silhouette analysis provided a clearer indication of the optimal number of clusters. The silhouette score measures how similar each observation is to the other observations within its assigned cluster compared with the observations in the nearest neighboring cluster. For an individual observation, the silhouette coefficient is calculated using Equation 7, where $a(i)$ is the average distance between observation i and the other observations in its assigned cluster, and $b(i)$ is the lowest average distance between observation i and the observations in another cluster. The coefficient ranges from -1 to 1, with higher values indicating that an observation is more similar to its own cluster and better separated from neighboring clusters (Rousseeuw, 1987). The average silhouette score is then calculated across all observations for each value of k. The average silhouette score reached its maximum value of 0.603 at $k = 5$, compared with 0.586 at $k = 4$, an improvement of 0.017. Although the difference was small, the silhouette analysis provided an explicit measure of cluster separation in addition to the elbow method's assessment of within-cluster variation. Since the silhouette analysis gave its highest score at $k = 5$, five clusters were selected for the final k-means analysis, which grouped stance steps based on similarity across the five standardized characteristics.

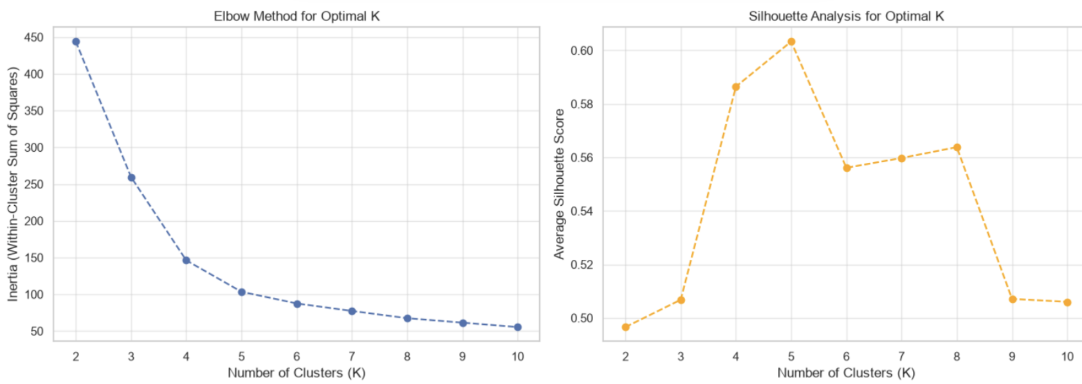

Figure 19: Elbow and Silhouette Analyses used to Determine the Number of Clusters for the k-means Analysis.

4 Results

After the filtered stance steps were converted into the five selected characteristics, the resulting feature matrix was standardized and clustered using k-means with five clusters. The clustering was performed using `n_clusters=5`, `n_init=10`, and `random_state=42`. A total of 177 steps were included in the final clustering, consisting of 40 randomly selected steps from LOG7, 38 steps from LOG11, 25 steps from LOG12, 37 steps from LOG13, and 37 steps from LOG14.

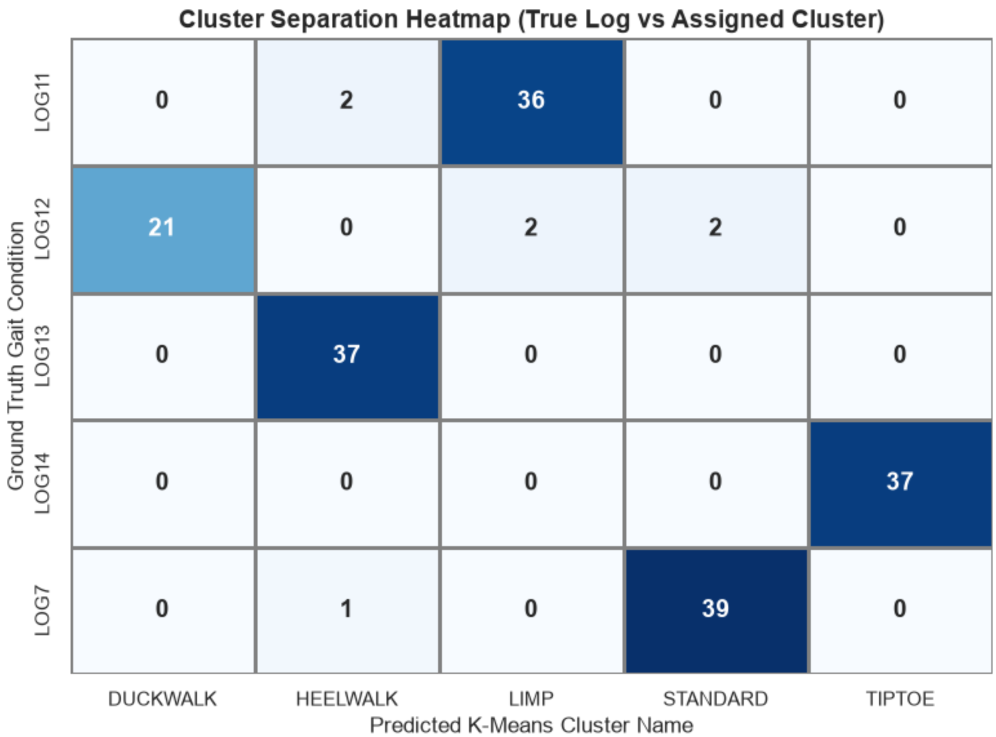

Figure 20: *k*-Means Clustering Heatmap

The resulting cluster assignments are shown in Table 1 and Figure 20. Each row represents one recorded gait condition, while each column represents the cluster assigned to the corresponding gait. LOG7 had 39 steps assigned to the STANDARD cluster and one step assigned to the HEELWALK cluster. LOG11 had 36 steps assigned to the LIMP cluster and two steps assigned to the HEELWALK cluster. LOG12 had 21 steps assigned to the DUCKWALK cluster, two steps assigned to the LIMP cluster, and two steps assigned to the STANDARD cluster. LOG13 had all 37 steps assigned to the HEELWALK cluster, and LOG14 had all 37 steps assigned to the TIPTOE cluster.

The agreement between the resulting clusters and the recorded gait labels was quantified using the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). The clustering produced an ARI of 0.9080 and an NMI of 0.9030.

Figure 21 shows the resulting clusters and their calculated centroids in the two-dimensional feature space. The five cluster centroids were subsequently associated with the STANDARD, HEELWALK, LIMP, DUCKWALK, and TIPTOE gait labels based on the recorded gait conditions

Table 1: Number of steps from each gait recording assigned to each *k*-means cluster.

Recorded Gait	DUCKWALK	HEELWALK	LIMP	STANDARD	TIPTOE
LOG7	0	1	0	39	0
LOG11	0	2	36	0	0
LOG12	21	0	2	2	0
LOG13	0	37	0	0	0
LOG14	0	0	0	0	37

represented within each cluster.

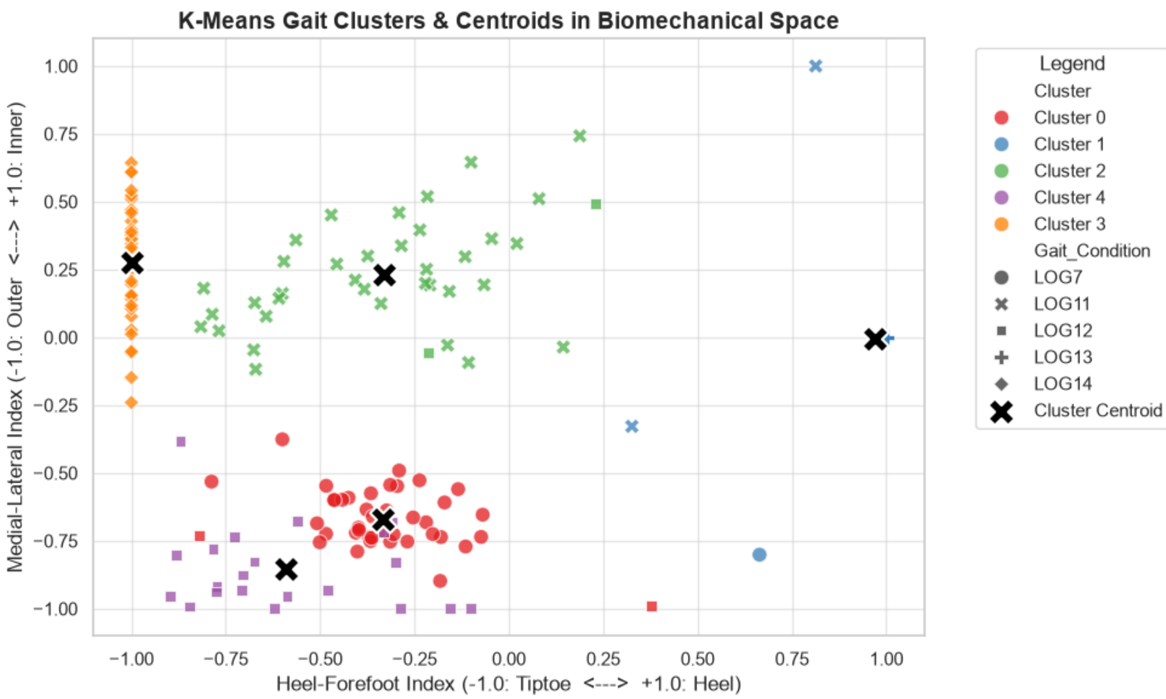

Figure 21: *k*-Means Clustering Scatterplot & Cluster Centroids

To visualize the five-dimensional feature space, principal component analysis (PCA) was applied to the standardized feature matrix. The first principal component accounted for 62.3% of the variance, while the second accounted for 25.9%, giving a combined variance of 88.1%. The resulting PCA projection and the calculated cluster centroids in the PCA-projected feature space is shown in Figure 22.

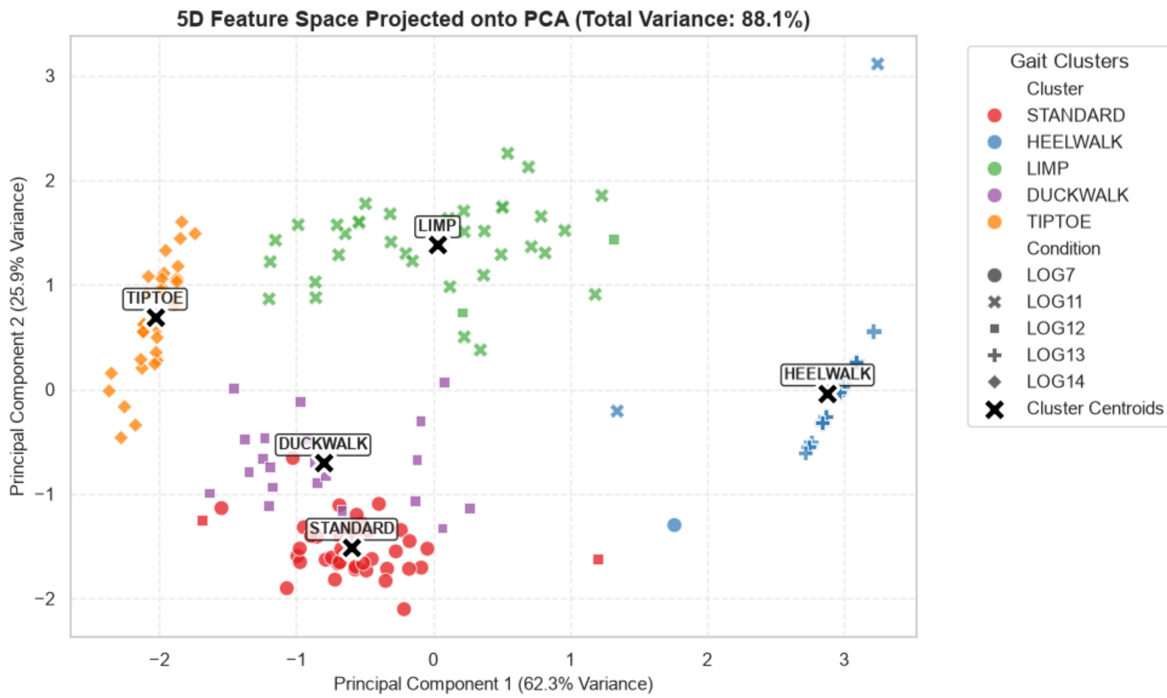

Figure 22: 2D Representation of 5D *k*-Means Clustering with Centroids

5 Discussion

5.1 Interpretation of Results

The clustering results showed that the five gait conditions were not equally distinguishable using the extracted temporal and plantar-pressure characteristics. Heel-walking and tiptoe gait were separated most consistently, with all 37 steps from LOG13 assigned to the HEELWALK cluster and all 37 steps from LOG14 assigned to the TIPTOE cluster. This separation is expected from the heel-forefoot index, which approaches $+1$ when pressure is concentrated on the heel and -1 when pressure is concentrated on the forefoot. The heel-walking condition maintained increased rearfoot loading throughout the gait cycle, producing a heel-forefoot index of 1.00 on average, while the tiptoe condition produced an average value of -1.00. These conditions therefore occupied opposite extremes of the heel-forefoot pressure distribution, providing a strong distinction for the clustering algorithm.

The remaining gait conditions showed greater overlap between clusters. LOG11, representing limp gait, had 36 of 38 steps assigned to the LIMP cluster, while LOG7, representing standard gait, had 39 of 40 sampled steps assigned to the STANDARD cluster. LOG12, representing duck-walk, had the greatest overlap, with 21 of 25 steps assigned to DUCKWALK and the remaining four steps assigned to LIMP or STANDARD. Unlike heel-walking and tiptoe gait, duck-walk is primarily characterized by a change in foot progression angle, meaning that the foot is rotated relative to the direction of travel. The fixed arrangement of six FSRs inside the insole measures pressure at specific locations on the foot, but does not directly measure the orientation of the foot relative to the direction of travel. Consequently, rotating the foot can produce pressure distributions that remain similar to those of other gait conditions, making duck-walk less distinct from the other clusters.

5.2 Limitations

The five characteristics used for clustering contain some redundancy. The heel-forefoot index is calculated from the same heel and forefoot pressure sums that are used to calculate the heel and forefoot percentages. Therefore, three of the five characteristics describe related aspects of the heel-to-forefoot pressure distribution. Because k-means is distance-based, correlated characteristics can give greater influence to this aspect of the pressure distribution than intended.

The choice of five clusters is also a limitation. The silhouette analysis produced its highest score at $k = 5$, but the score was only 0.017 higher than the score at $k = 4$, while the elbow method indicated a possible elbow around $k = 3$–4. Since five gait types were collected for the experiment, selecting five clusters also agreed with the expected number of gait categories. Therefore, the clustering result should not be interpreted as definitive evidence that five is the uniquely correct number of gait clusters.

The silhouette score may also be optimistically biased by the filtering procedure. Each LOG was filtered using thresholds calculated from its own step-length distribution before the clustering features were evaluated. This removed unusually long and short detections within each recording and may have reduced within-LOG variation before the silhouette score was calculated. As a result, the silhouette score of 0.603 may represent a more separated dataset than would have been obtained without the per-LOG filtering.

Subsampling also affected the clustering dataset. LOG7 originally contained 850 filtered steps, but only 40 were randomly selected for clustering so that the Standard Gait recording did not disproportionately influence the *k*-means model. This produced a more balanced dataset across the five gait recordings, but discarded most of the available Standard Gait data. The 40 selected steps may therefore not fully represent the variation present throughout the approximately 17 minute recording.

The experiment was performed using a single subject measuring 183 centimeters and weighing 56.83 kilograms. In addition, the four abnormal gait patterns were simulated by the subject rather than collected from individuals experiencing the corresponding conditions. The results therefore demonstrate that the Insole and processing method can distinguish the simulated gait patterns under the tested conditions, but they cannot establish that the method will generalize to different subjects or naturally occurring pathological gait.

The Insole recorded data at approximately 24Hz, providing roughly 15 samples during a typical stance phase. This sampling rate limits the temporal resolution of the pressure measurements and may miss short-duration changes in plantar pressure that would be captured by a higher sampling rate. Consequently, the extracted step characteristics represent relatively coarse measurements of the pressure distribution throughout each stance phase.

The step segmentation method also has limitations because it defines the swing phase using rows where all six FSR readings are zero. Baseline drift or sensor noise could cause a sensor to report a nonzero value during the swing phase, while a temporary zero reading from all sensors could divide a single stance phase into multiple segments. Therefore, the accuracy of the detected steps depends on the sensors consistently returning zero when the foot is not applying pressure.

Finally, the filtering thresholds of 0.2 standard deviations for long steps and 2.9 standard deviations for short steps were selected through inspection of the recorded data rather than from an independently established criterion. The filtering was relatively insensitive to the selected threshold for LOG11 to LOG14, but LOG7 had a much narrower separation between normal and removed steps. The selected thresholds could be less reliable for recordings with greater variation in stance-step duration.

5.3 Future Work

Despite

6 Conclusion

This study used unsupervised *k*-means clustering to classify five recorded gait conditions using five step-level characteristics extracted from six FSRs in a pressure-sensing insole. The clustering results achieved an Adjusted Rand Index (ARI) of 0.9080 and correctly assigned 170 of the 177 analyzed steps to their corresponding gait clusters.

These results indicate that the extracted temporal and plantar-pressure characteristics contained sufficient information to distinguish the five gait conditions recorded in this experiment. The results are limited to the simulated gait conditions and experimental setup used in this study and do not establish that the method can identify clinical gait abnormalities in patients.

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